

A shared item representation forces a three-way trade-off in neural IRT
the gradient flow predicts it; the experiments confirm it

Wenrui Yuan

Setup: amortized neural IRT

- Encoder (LSTM): response history $\rightarrow$ ability θ_t
- Item embedding $\rightarrow$ discrimination α_i , difficulty (step thresholds) β_i ; GPCM decoder

$$P(Y_i = k \mid \theta) = \frac{\exp \sum_{m=1}^k \alpha_i(\theta - \beta_{i,m})}{\sum_{l=0}^{K-1} \exp \sum_{m=1}^l \alpha_i(\theta - \beta_{i,m})} \quad (2\text{PL at } K=2)$$

- Trained on response prediction only. No supervision on θ, α, β .
- Scale is gauge-free $\Rightarrow$ recovery = rank (Spearman ρ).

The item code has two consumers

- a **value** embedding v_q (narrow) $\rightarrow$ encoder input $\rightarrow$ **ability** θ
 - a **key** embedding k_q (wide) $\rightarrow$ **discrimination** α , **difficulty** β readouts
 - **Shared**: one embedding, width W , feeds both consumers
 - **Decoupled**: a narrow value embedding + a separate wide key embedding
- $\Rightarrow$ *shared ties the encoder-input width to the readout width.* That tie is the problem.

First the math: the gradient flow of all three

Per-parameter gradients, residual $r = p - y$:

$$\partial_{\theta} L = r\alpha, \quad \partial_{\beta} L = -r\alpha, \quad \partial_{\alpha} L = r(\theta - \beta).$$

Near the optimum each parameter's mode decays at its Fisher rate ($w = p(1-p)$):

$$\underbrace{I(\theta) = I(\beta) = \alpha^2 w}_{\text{fast}}, \quad \underbrace{I(\alpha) = (\theta - \beta)^2 w}_{\text{slow, } \rightarrow 0 \text{ at } \theta = \beta}.$$

Discrimination is the one low-rate mode: its lever arm $(\theta - \beta)$ vanishes exactly where responses concentrate. **The math singles out α before any experiment.**

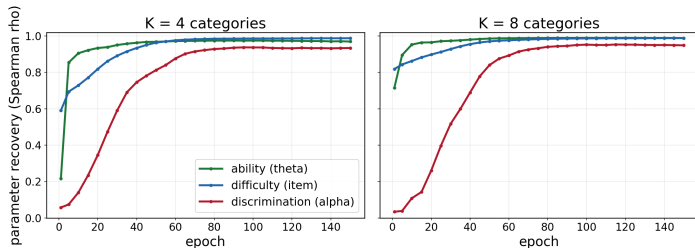
What the math predicts (then we test each)

- ① **Discrimination recovers last** (it is the single slow mode; θ, β fast).
- ② **A shared code forces a three-way trade-off**: θ wants it narrow, α wants it wide, β is indifferent.
- ③ **Sharing adds conditioning** $\kappa = I(\theta)/I(\alpha)$: it throttles α 's rate; decoupling removes it (speedup $\sim \kappa$, growing with K).
- ④ **Positivity is necessary, the map shape is not**: among smooth positive maps the convergence is the same; exp is not special.

Synthetic GPCM, known θ, α, β , LSTM. Each prediction is tested next.

Prediction 1: discrimination recovers last

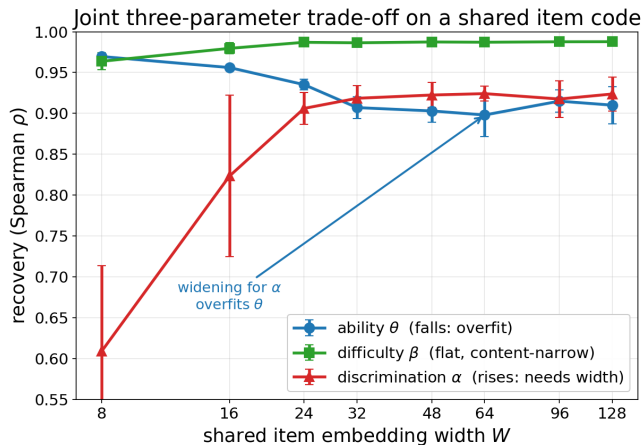
Ability and difficulty are learned fast; discrimination lags



- ability, difficulty: fast (high Fisher rate)
- discrimination: slow, lags by hundreds of epochs

Exactly the ordering the rates predict.

Prediction 2: the three-way trade-off



As the shared code widens:

- α rises 0.61 $\rightarrow$ 0.92 (gets capacity)
- θ falls (overfits the wide input)
- β flat-high 0.96 $\rightarrow$ 0.99

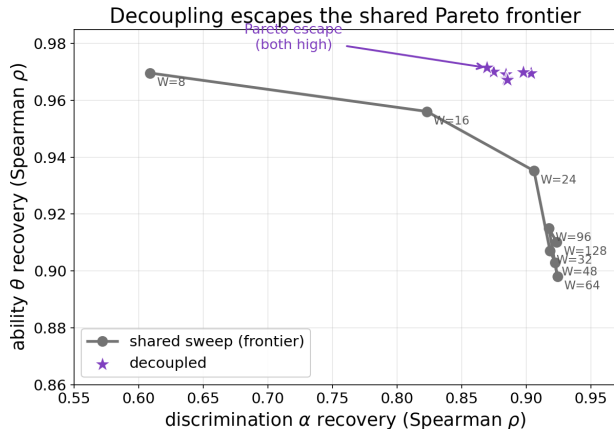
One knob, opposed pressures. β does not pay.

The two pressures are of different KIND

- α : **a RATE effect, endpoint invariant.** α is a free per-item table, so the optimum is unbiased (free-table invariant). Sharing only slows it (conditioning κ); decoupling speeds it. Transient: vanishes as training $\rightarrow \infty$, not at a fixed budget. (P9)
- θ : **an ENDPOINT effect, not invariant.** θ is read by the amortized encoder, NOT a free table. A wide input overfits it, $\text{Var}(\hat{\theta}) \sim \sigma^2 W/n$. It does NOT vanish at convergence. (P10)
- β : **indifferent control.** High Fisher, non-pooled $\Rightarrow$ neither needs width nor is hurt ($\Delta_\beta \approx 0$). (P11)

A slow mode (rate) *and* an overfit (endpoint), tied by one width.

Prediction 3a: not capacity, allocation

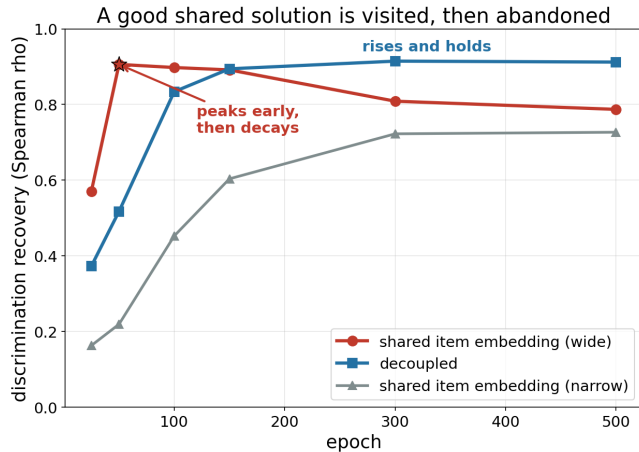


Control: matched total item-embedding params.

- widening the shared code traces a Pareto frontier
- decoupled sits **above** it, even at $6\times$ the shared budget

$\Rightarrow$ allocation, not budget. Decoupling escapes the frontier (P12).

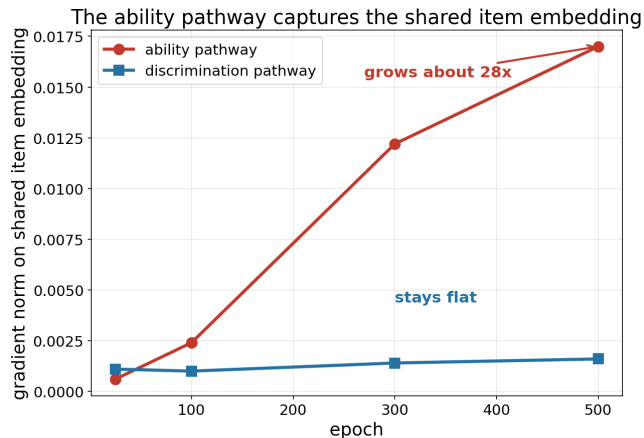
Prediction 3b: the good α solution is reached then left



- wide-shared α : peaks 0.91 (ep 50), decays to 0.79 (ep 500)
- decoupled: rises 0.91, holds

$\Rightarrow$ reachable, reached, then abandoned
= dynamics, not a ceiling.

Prediction 3c: the shared code is conditioned



- θ -pathway pull grows $\sim 28\times$;
 α -pathway flat
- $\cos(g_\theta, g_\alpha) \approx 0$ (not a tug-of-war)
- the lag is eigenvalue spread,
stiffness $\kappa \approx 1.9$ at $K=4$

Prediction 3: the gradient-flow ODE

One item's embedding e_q . Near its optimum, $\delta = e_q - e_q^*$, H_q the Hessian in e_q 's directions:

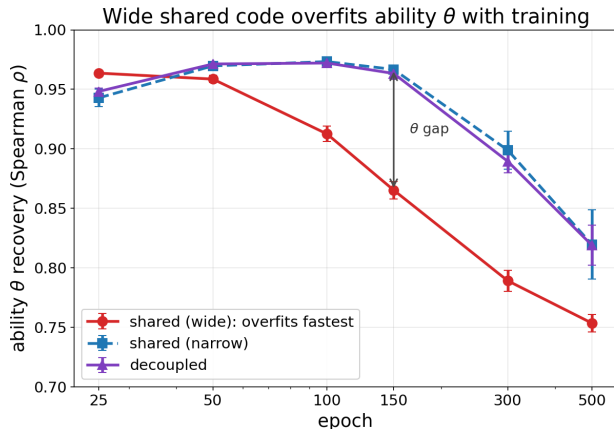
$$\dot{\delta} = -H_q \delta, \quad \delta_c(t) = \delta_c(0) e^{-\lambda_c t}.$$

- Shared: e_q feeds encoder (θ) and α, β readouts $\Rightarrow$ steep θ direction ($\sim I(\theta)$), flat α direction ($\sim I(\alpha)$): $\kappa = I(\theta)/I(\alpha) \gg 1$
- one step $\eta < 2/\lambda_{\max}$: slow mode $\sim 1/\kappa \Rightarrow \tau_{\alpha}^{\text{sh}} \sim O(\kappa)$; decoupled $O(1)$

speedup $\sim \kappa = I(\theta)/I(\alpha)$ (grows with K)

Same fixed point $\Rightarrow$ rate, not endpoint. Largest under GD; Adam compresses κ .

The θ side: an endpoint overfit



- wide-shared θ overfits worst: $0.96 \rightarrow 0.75$
- decoupling keeps the encoder narrow $\Rightarrow$ protects θ at the operating point

Bias and variance: $\text{Var}(\hat{\theta}) \sim \sigma^2 W/n$. An **endpoint** loss (not a rate), outside the free-table invariant. (The decay with long training, even narrow, is a separate regularization issue, not the width effect.)

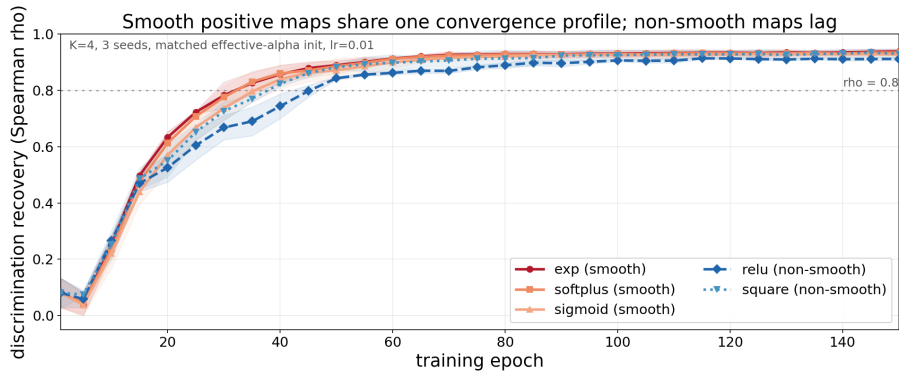
The resolution: untie the two widths

- **Shared:** one width, opposed knobs ($d\alpha/dW > 0$, $d\theta/dW < 0$, $d\beta/dW \approx 0$) $\Rightarrow$ a Pareto frontier
- **Decoupled:** two independent widths $\Rightarrow$ narrow encoder (good θ, β) + wide readout (good α)

$\Rightarrow$ decoupling is Pareto-**dominant**: it escapes the frontier, not moves along it. (P12)

It cures both pressures at once: θ keeps its bottleneck, α gets its capacity.

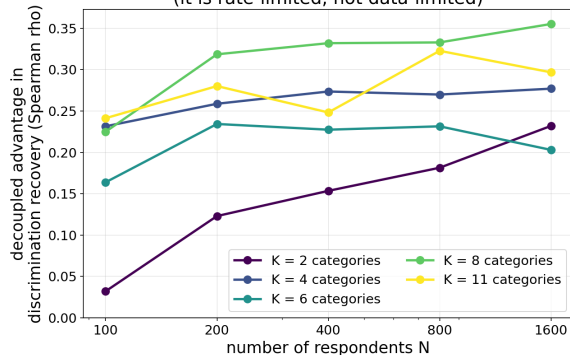
Prediction 4: positivity needed, map shape is not



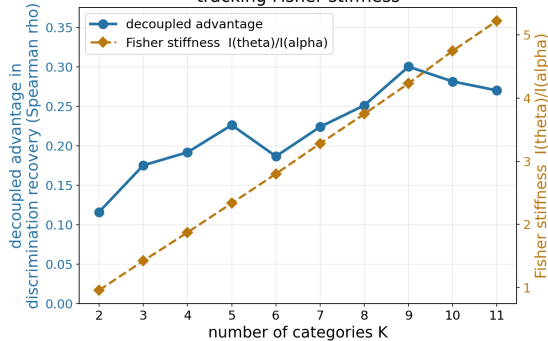
- smooth maps (exp, softplus, sigmoid) share one convergence profile; non-smooth (relu, square) lag mid-training
- endpoints nearly tie, so the *profile* separates them, exp is not special
- direct- α control (θ, β frozen): a scalar reparameterization does nothing ($\kappa=1$) $\Rightarrow$ the lever is the representation, not the map

The α rate effect: finite-sample and K -scaled

More data does not shrink the advantage
(it is rate-limited, not data-limited)



The advantage grows with the number of categories K ,
tracking Fisher stiffness



- Fixed budget, vary N : the decoupling advantage does NOT shrink with data (rate-limited).
- Across $K=2$ to 11: it tracks the Fisher stiffness κ , $\rho = 0.89$.

Conclusion

The gradient flow predicts, and the experiments confirm, that a shared item code forces a three-way trade-off: θ overfits wide, α needs wide, β is content-narrow. Decoupling unties the widths and dominates.

- Two mechanisms of different kind: α a convergence-RATE effect (endpoint invariant, $O(\kappa)$, grows with K); θ an ENDPOINT overfit ($\text{Var} \sim W/n$). β indifferent.
- A measurement-validity condition: recovered parameters are otherwise representation-allocation artifacts.
- Positivity map ruled out as the lever.

Scope: synthetic GPCM, LSTM (architecture-independent on swap), 3 to 10 seeds. θ -overfit formalized for a linear amortizer (LSTM lift argued). Open: NRM slopes, GRM, real data.